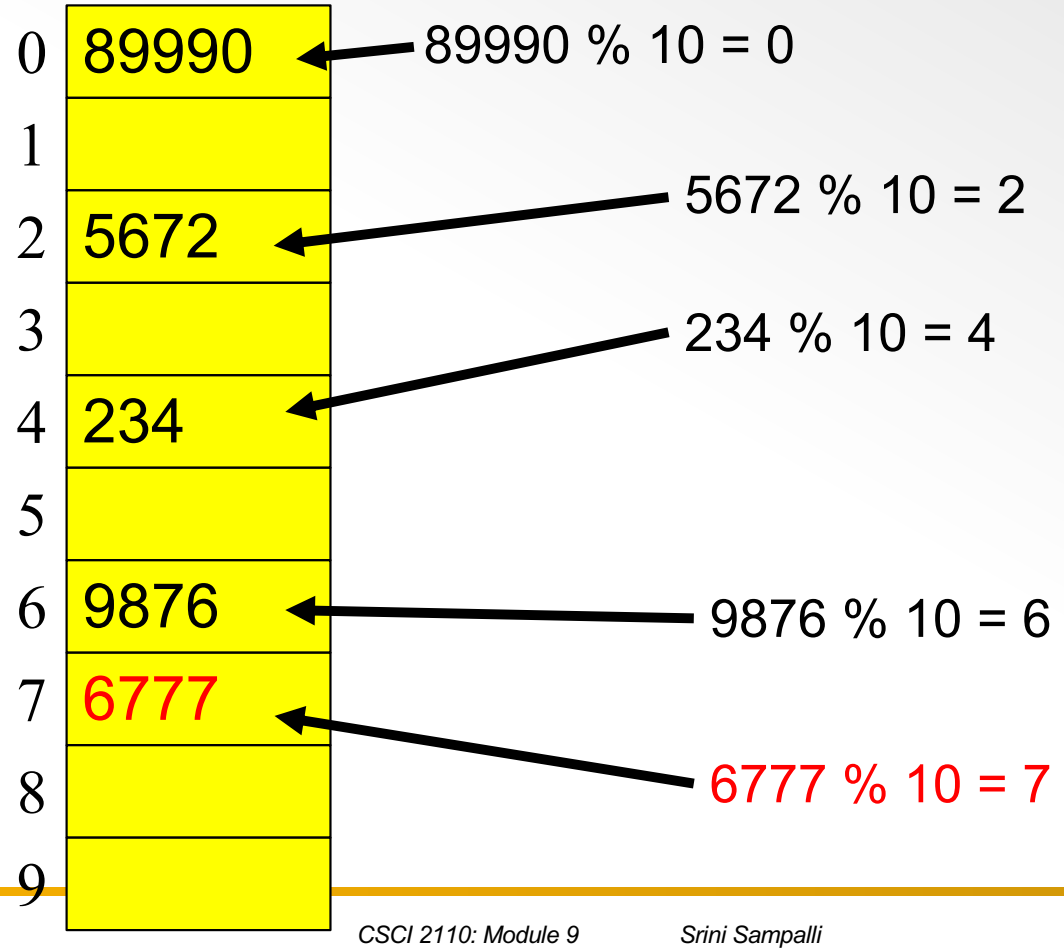
Example

Keys:
9876, 234, 5672, 89990, 6777



Now look at the search operations!

0	89990
1	
2	5672
3	
4	234
5	
6	9876
7	6777
8	
9	

Now look at the search operations!

0	89990
1	
2	5672
3	
4	234
5	
6	9876
7	6777
8	
9	

Search for key 234
Apply the hash function
234 %10 = 4
Go to location 4
Found!
O(1) time complexity!

Now look at the search operations!

0	89990
1	
2	5672
3	
4	234
5	
6	9876
7	6777
8	
9	

Search for key 234
Apply the hash function
234 %10 = 4
Go to location 4
Found!
O(1) time complexity!

Search for key 878
Apply the hash function
878 %10 = 8
Go to location 8
Not found!
O(1) time complexity!

Now look at the search operations!

0	89990
1	
2	5672
3	
4	234
5	
6	9876
7	6777
8	
9	

Search for key 234
Apply the hash function
234 %10 = 4
Go to location 4
Found!
O(1) time complexity!

Search for key 878
Apply the hash function
878 %10 = 8
Go to location 8
Not found!
O(1) time complexity!

Search for key 312
Apply the hash function
312 %10 = 2
Go to location 2
Not found!
O(1) time complexity!

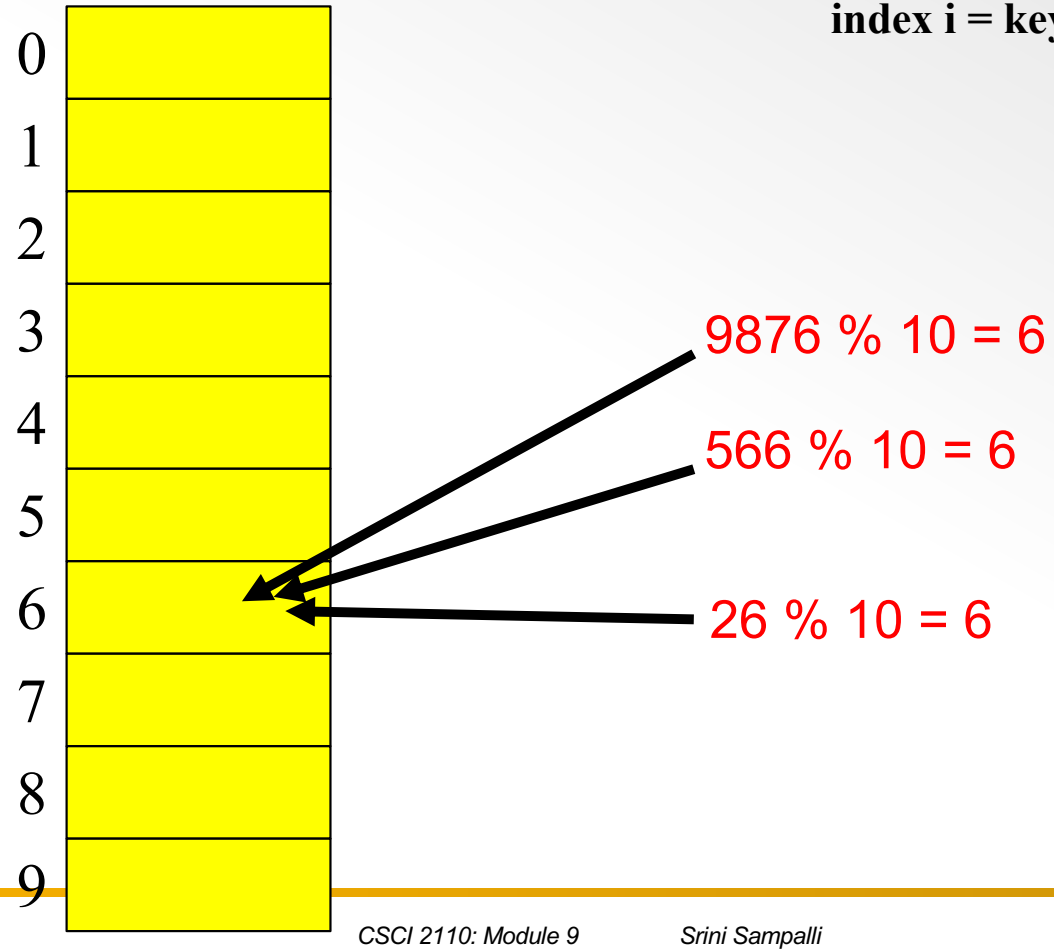
Problem:
What if two or more keys map to the same index?

This is bound to happen since we are mapping a large range of keys into a small table.

Example

Keys:
9876, 566 and 26

index i = key % 10



The Hash Clash Problem

If two or more keys map to the same index, it is called a Hash Clash or a Hash Collision.

The Hash Clash Problem

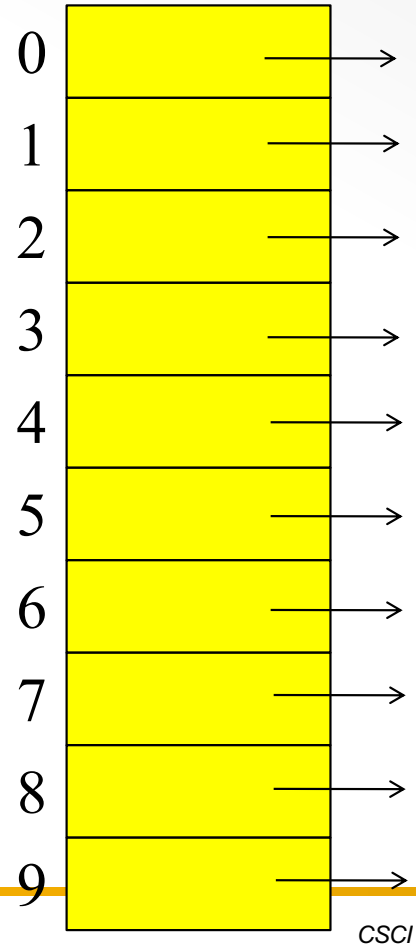
If two or more keys map to the same index, it is called a Hash Clash or a Hash Collision.

<u>Two Solutions:</u>

- ***OPEN HASHING (SEPARATE CHAINING):***
- ***CLOSED HASHING***

Open Hashing or Separate chaining

Concept: Create an ArrayList of Linked Lists

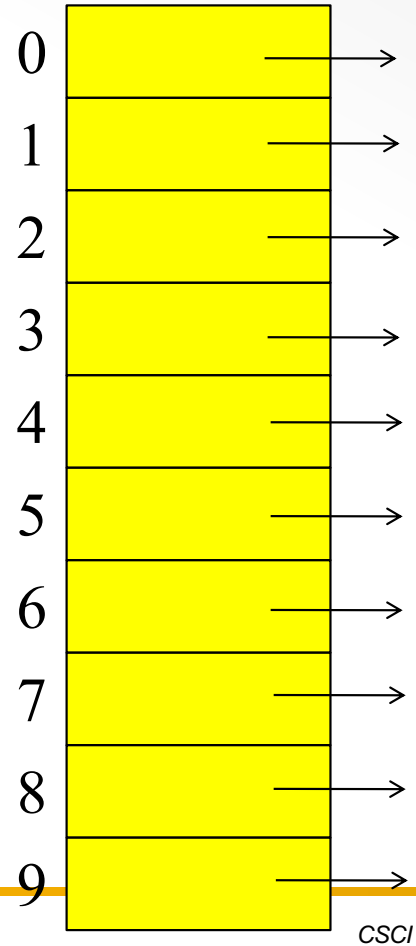


Keys:

9876, 234, 5672, 89990, 6777, 366, 7446, 212, 4676

Open Hashing or Separate chaining

Concept: Create an ArrayList of Linked Lists

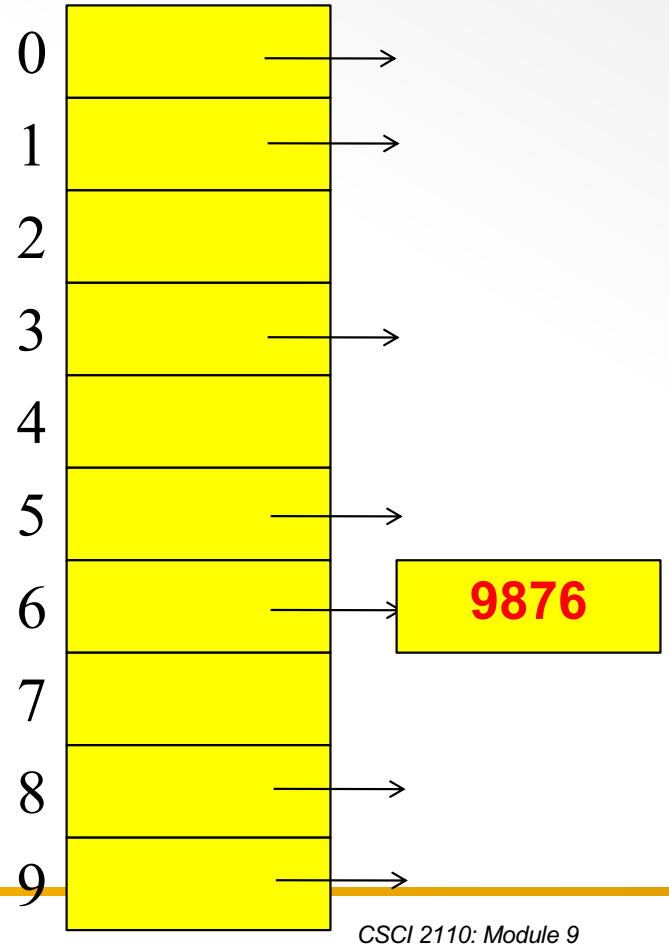


Keys:

9876, 234, 5672, 89990, 6777, 366, 7446, 212, 4676

Open Hashing or Separate chaining

Concept: Create an ArrayList of Linked Lists



Keys:

9876, 234, 5672, 89990, 6777, 366, 7446, 212, 4676